

BILLAL HOSSAIN SHISHIR

Information Technology | Artificial Intelligence, Data & Software

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PROFESSIONAL SUMMARY

Artificial Intelligence undergraduate at the University of Tasmania with hands-on experience building and deploying full-stack AI applications, machine-learning pipelines, cloud data workflows and algorithms in C. Built a live AI-assisted IT service desk with FastAPI, PostgreSQL and scikit-learn, including explainable classification, information retrieval, automated testing and deployment. Project Manager for RAMIFY OS, coordinating requirements, testing, documentation and client feedback. Brings 2+ years of customer service experience and clear stakeholder communication.

CORE TECHNICAL SKILLS

Programming & Web: Python, C, JavaScript, HTML, CSS, SQL, Chart.js

AI & Data: pandas, NumPy, scikit-learn, text classification, TF-IDF, Logistic Regression, cosine similarity, information retrieval

Algorithms & Data Structures: Arrays, linked lists, AVL trees, hash tables, weighted graphs, Dijkstra shortest path, binary-heap priority queues

Backend & Databases: FastAPI, PostgreSQL, SQLAlchemy, Pydantic, SQLite, REST APIs, Uvicorn

Cloud, DevOps & Testing: Docker, GitHub Actions, Render, pytest, Git/GitHub, Nectar Cloud, AWS fundamentals, Linux, Windows

Project & Delivery: Requirements analysis, functional testing, documentation, stakeholder communication, presentations, project coordination

PROFESSIONAL EXPERIENCE

Project Manager - RAMIFY OS | University Industry Project | Hobart, TAS

Jul 2026 to Present

- Lead project planning and coordination for a browser-based AI product-trust demonstration that produces auditable decision receipts from structured product evidence.
- Coordinate project scope, requirements, team responsibilities, testing, documentation and client feedback while supporting mandatory human-review workflows.
- Manage deliverables including architecture material, user guidance, scripted demonstrations and final presentation preparation across multiple trust scenarios.

Customer Service Representative | Bennett's Petroleum | Hobart, TAS

Oct 2024 to Present

- Provide responsive customer service, handle transactions and resolve enquiries in a fast-paced environment while maintaining accuracy and clear communication.
- Complete day-to-day operational tasks, identify recurring issues and contribute practical solutions while working independently and as part of a team.

Crew Member | McDonald's Rosny Park | Hobart, TAS

Jun 2024 to Sep 2024

- Processed customer orders and transactions accurately, coordinated with kitchen colleagues and maintained organised work areas during busy service periods.

Customer Service Assistant | Gadget and Gear | Dhaka, Bangladesh

Apr 2023 to Dec 2023

- Assisted customers with product enquiries, recommendations, payments, returns and complaint resolution while supporting stock replenishment and inventory checks.

SELECTED PROJECTS

- AI Helpdesk Copilot - Live Full-Stack AI Application

Sep 2026

Live Demo: ai-helpdesk-copilot.onrender.com | GitHub: github.com/Billalhossainshishir/ai-helpdesk-copilot

 - Built and deployed an AI-assisted IT service desk using Python, FastAPI, PostgreSQL, SQLAlchemy, scikit-learn and JavaScript, with persistent ticket data and interactive API documentation.
 - Developed a TF-IDF and Logistic Regression classifier for seven IT support categories using 490 synthetic, portfolio-safe training examples, paired with transparent priority rules.
 - Implemented TF-IDF cosine-similarity retrieval across 70 troubleshooting articles and previously resolved incidents, keeping recommendations traceable to matched support evidence.
 - Delivered an end-to-end workflow covering issue analysis, ticket creation, assignment, technician notes, status and resolution tracking, similar-incident search and operational analytics.
 - Added 25 automated tests with pytest and FastAPI TestClient, GitHub Actions for repository changes, and deployment on Render with PostgreSQL persistence and Docker support.
- Cloud Data Processing & Machine Learning

Feb 2026 to Jul 2026

 - Served as Project Manager and built/tested Worker 3 in a distributed Nectar Cloud workflow, encoding categorical data and preparing numerical features for machine learning.
 - Compared Logistic Regression, Decision Tree and Random Forest on 1,221 records using Python, pandas and scikit-learn; applied cross-validation and GridSearchCV, with the best tuned model reaching approximately 0.814 F1.
- Data Structures & Algorithms Projects

Feb 2026 to Jul 2026

 - Built and benchmarked a C food-delivery database prototype using arrays, AVL trees, hash tables and linked lists; AVL trees performed best overall in the evaluation.
 - Implemented emergency route planning in C using weighted graphs, Dijkstra shortest path and a binary-heap priority queue; 24 unit tests covered graph operations, route reconstruction and invalid or unreachable cases, with all tests passing.
- Star Type Classification with Machine Learning

Sep 2026

 - Built a supervised multiclass Python workflow to classify six star types, including data-quality review and controls to avoid target leakage.
 - Compared Decision Tree, Random Forest and SVC with stratified 5-fold cross-validation, then tuned Random Forest with GridSearchCV to 0.9947 mean cross-validation accuracy.

INDUSTRY & RESEARCH ENGAGEMENT

- Open Innovation Hackathon | TasNetworks x University of Tasmania x AWS | Hobart, TAS

May 2026

 - Collaborated over three days to design and present a prototype solution for a real-world Tasmanian energy-sector challenge, applying teamwork, rapid problem-solving and AI/cloud concepts.
- DataEngBytes Melbourne 2026 | Industry Conference | Melbourne, VIC

23-24 Jul 2026

 - Attended industry sessions covering AI engineering, GraphRAG, Snowflake, synthetic data, observability, data architecture and agent workflows.

EDUCATION

- Bachelor of Information Technology - Major in Artificial Intelligence | University of Tasmania

Feb 2024 to Expected Feb 2027

Hobart, TAS
- International First Year Diploma Science | University of Tasmania

Feb 2024 to Feb 2025

ADDITIONAL

Location: West Hobart, Tasmania
Registration: Registration to Work with Vulnerable People (Tasmania) | No. 716007141
References: Available on request